EPSR-test-ex53 triangles: bond-angle distributions

Program(s): triangles | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises `triangles`, which computes distributions of the angle atom1-atom2-atom3 for atom pairs within distance windows [1], either between real atoms (`ltype 1`) or with the third point the mean position of the atom3 atoms bonded to atom2 (`ltype 2`), optionally restricted by the distance from a centre type (`atom-c`, `atom-z`, `atom-r`), and an orientational order parameter per distribution.

Code exercised

Fortran (original, fortran_src/):

- `triangles.f`: `run_triangles` (normalisation, order parameter), `find_triangles`
- gfortran intrinsic `SIND`/`COSD` (`_gfortran_sind_r4`, `_gfortran_cosd_r4`)
- `read_and_save_files.f`: `read_input_file`, `initgofz`, `outgofz` (histogram file, running average and spread), `save_files_others`; `auxiliary_routines.f`: `r_int`
- `setup_symbols_generic.f`: `setup_symbols_triangles`

Rust (port, rust/epsr-core/src/):

- `tools/triangles.rs`: `run`, `find_triangles`
- `fortran/fmath.rs`: `sind_intrinsic`, `cosd_intrinsic` (re-exported by `tools/anal_common.rs`)
- `tools/anal_common.rs`: `match_centre`, `outgofz`; `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box (200 DMSO with atom types S, O, C, H and 400 water molecules with O1, H1; 3200 atoms, 300 K, cubic box 33.1213 Å, 0.0881 atoms/Å³), the EPSR model of DMSO-water neutron data [2]. Each distribution is normalised by sin(theta) to unit area. `water.TRI.dat`: O1-O1-O1 angles between neighbouring water oxygens (O-O 2.0-3.4 Å), the signature of the tetrahedral hydrogen-bond network; O1-O1-(H1 of the middle O1); S-O1-H1 for water within 2.5-4.5 Å of S. For `ltype 1` the order parameter is q = 1 - (9/4)<(cos theta + 1/3)²>, equal to 1 for perfect tetrahedra; the reference gives q = 0.490 for O1-O1-O1. `slabz`, `cylr` and `centreS` restrict the triplets by their z, xy or 3D distance from DMSO O or S.

Inputs

Four keyword files `slabz`, `cylr`, `centreS`, `water .TRI.dat` (2, 2, 2 and 3 distributions; `slabz` with `nsumt -1`, the others `nsumt 0`); `200DMSO400Water.ato` copied unchanged from EPSR-test.

How the accuracy was verified

`commands.txt` (4 program calls, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
The four commands run in sequence, the list twice. Compared: the four `.TRI.c01` files, the rewritten `.dat` files and `fort.14` (list-directed order parameters) of the last run (`water`); the README states that `fort.14` of the other three inputs was also checked identical in separate runs. The configuration is the same in every pass, so the chained passes test the read-back of the accumulated file and the running-average arithmetic.

Notes

`triangles.f` calls `sind`/`cosd` without declaring them EXTERNAL, so gfortran 15 uses its intrinsic SIND/COSD, not the routines of `auxiliary_routines.f`; the port reproduces the libgfortran algorithm bit for bit (unit test `trig_intrinsics_match_gfortran`).

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.